WAVE & OPTICS
MECHANICAL WAVES

Theoretical Questions

1. **Discuss the effect of temperature and pressure on velocity of sound in a gas.**
   Ans: Effect of Temperature:
   Velocity of sound increases with temperature as higher temperature increases molecular kinetic energy. i.e.
   $$V \propto T$$
   For every 1C rise, speed increases by about 0.6 m/s in air.
   Effect of Pressure:
   At constant temperature, pressure has no effect on velocity because both pressure and density increase proportionally, keeping (P\ρ) constant.
   Note: Also check the notes of effect of temperature.

2. **Explain the effect of humidity on the velocity of sound in air.**
   Ans: The amount of water vapor present in air is called humid air or humidity. The molecular mass or density of humid air is less than that of normal or dry air.
   Since, velocity of sound is inversely proportional to square root of molecular mass or density. This shows that velocity of sound increases when humidity increases.

3. **Explain why sound waves are also called pressure waves.**
   Ans: Sound waves are called pressure waves because they travel through a medium by causing alternating compressions and rarefactions, which create variations in pressure.

4. **Discuss the effect of density of elastic medium on velocity of sound.**
   Ans: The velocity of sound is inversely proportional to the square root of the density of the medium:
   $$\text{Velocity (V)} = \sqrt{\frac{E}{p}}$$
   Where,
   V = velocity of sound
   E = elastic modulus (elasticity)
   P = density of the medium
   This means that if the density increases, the velocity of sound decreases, provided the elasticity remains constant.

5. **What are the assumptions of Newton and Laplace on deriving the formula for velocity of sound in gas? Explain.**
   Ans: Newton's Assumption:
   Newton assumed that sound waves in a gas travel through an isothermal process, meaning the temperature remains constant during compression and rarefaction. Based on this, he derived:
   $$(V) = \sqrt{\frac{P}{p}}$$
   However, this value was less than the actual observed speed.
   Laplace's Correction:

Laplace corrected Newton's assumption by stating that sound waves travel very fast, so no heat is exchanged - the process is adiabatic.
He modified the formula to:

$$(V) = \sqrt{\frac{\gamma P}{p}}$$

Where γ is the adiabatic index. This gave results that matched experimental values.

Short Answer Questions

6. **Velocity of sound in solids is more than that in liquids, why?**
   Ans: The velocity of sound depends on:

   $$\text{Velocity (V)} = \sqrt{\frac{E}{p}}$$

   In solids, the elastic modulus (E) is much higher than in liquids, meaning solids are more rigid and resist deformation better.
   Although solids are denser, their high elasticity dominates, resulting in a higher speed of sound compared to liquids.

7. **The velocity of sound in solid is generally greater than that in gas at STP. Why? Give reason.**
   Ans: The velocity of sound is given by:

   $$\text{Velocity (V)} = \sqrt{\frac{E}{p}}$$

   Solids have a much higher elastic modulus (E) compared to gases, meaning they are more rigid and transmit vibrations more effectively.
   Even though solids are denser, their greater elasticity results in a much higher sound velocity than in gases at STP.

8. **When sound waves travel through a medium, does the temperature at various points remain constant? Explain.**
   Ans: No, the temperature does not remain constant at all points.
   Sound waves cause compressions and rarefactions, leading to local changes in pressure and volume.
   Since these changes happen rapidly, the process is adiabatic, and temperature fluctuates slightly at different points — it increases during compression and decreases during rarefaction.

   Note: *In an adiabatic process, no heat is given to or taken from the surroundings. The temperature changes happen only due to compression or expansion of the gas.*
   👉 *When a gas is compressed (squeezed), the molecules get packed closer, move faster, and the temperature increases.*
   👉 *When a gas expands (released), the molecules spread out, move slower, and the temperature decreases.*
   This is an internal change — the heat is not going out, and heat is not coming in. All temperature change happens just because of the work done — not because of external heat.

9. **Why does sound travel faster in metals than in air?**
   Ans: Sound travels faster in metals because metals have a very high elastic modulus compared to air even though metals are denser.

10. **The speed of sound in humid air is more than that in dry air, why?**
    Ans: The density of humid air is less than that of normal or dry air. Since, velocity of sound in inversely proportional to square root of density. Therefore, the speed of sound in humid air is more than that in dry air.

11. **Although the density of solid is high, the velocity of sound is greater in solid, explain.**
    Ans: The velocity of sound is given by:

    $$\text{Velocity (V)} = \sqrt{\frac{E}{p}}$$

    Where E is the elastic modulus and ρ is density.
    Though solids have high density, they also have a much higher elastic modulus, which increases the numerator significantly. This high elasticity overcomes the effect of density, making the velocity of sound greater in solids.

12. **Sound at a distance can be heard distinctly at night than in the day time. Why?**
    Ans: At night, the ground cools and the upper air stays warm. Sound moves slower near the ground and faster above, so it bends downward due to refraction, staying close to the surface.
    Also, humidity at night is higher, reducing air density, helping sound travel faster.
    Thus, sound is heard more clearly at night due to refraction and humidity.

13. **Do sound waves need a medium to travel from one point to other point in space? What properties of the medium are relevant?**
    Ans: Yes, sound waves need a material medium to travel, as they are mechanical waves that propagate by vibrating particles.
    Key properties of the medium:
    Elasticity: Greater elasticity helps particles restore quickly, increasing speed.
    Density: Higher density generally slows sound if elasticity is not high.
    So, elasticity and density of the medium mainly affect the velocity and clarity of sound.
    Sound cannot travel in a vacuum.

Short Answer Questions

14. **Discuss Newton's formula for the velocity of sound in air and explain why and how this formula was modified by Laplace.**
    Ans: Newton's Formula:
    Newton assumed that sound waves travel in air under isothermal conditions (i.e., temperature remains constant). He derived the velocity of sound as:

    $$(V) = \sqrt{\frac{P}{p}}$$

    Where:
    V = velocity of sound
    P = pressure of the gas

ρ = density of the gas

But this gave a value 15% lower than the actual speed of sound in air. Laplace pointed out that sound travels very rapidly, not allowing time for heat exchange. So, the process is adiabatic, not isothermal.

He corrected the formula as:

$$(V) = \sqrt{\frac{\gamma P}{p}}$$

Where γ is the adiabatic index.

Note: Check derivation in the notes.

15. Does the propagation of sound wave cause change in thermodynamic condition of medium?

Ans: Yes, sound wave propagation causes rapid compressions and rarefactions, leading to small changes in pressure, volume, and temperature. These changes occur adiabatically (without heat exchange), thus altering the thermodynamic state of the medium.